Maximum Extractable Value (MEV) Mitigation Approaches in Ethereum and Layer-2 Chains: A Comprehensive Survey

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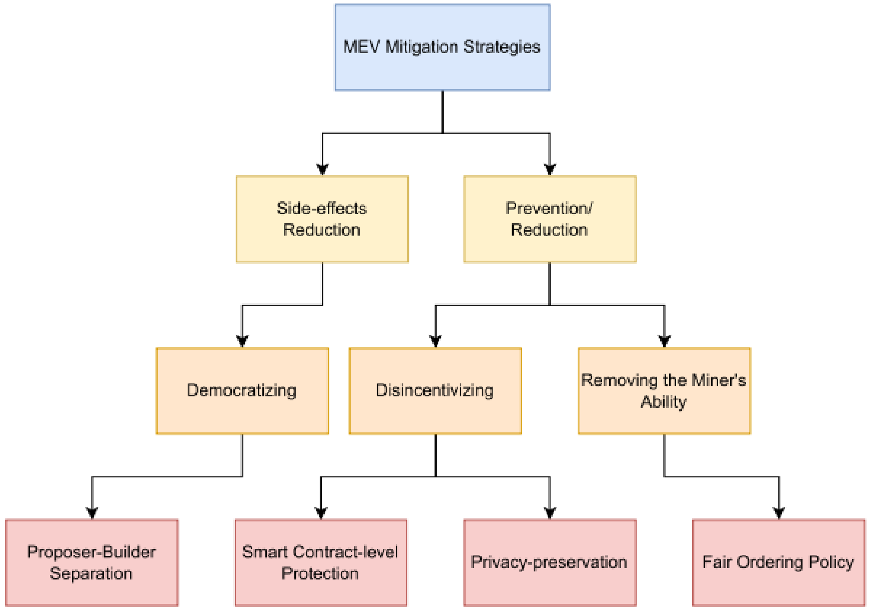
MITIGATION STRATEGIES

Two major categories:

Prevention/Reduction: try to eliminate or reduce MEV opportunities.

Side-effects Reduction: try to reduce the negative effects.



PREVENTION/REDUCTION: FAIR ORDERING POLICIES

If the problem arise because sequencer can reorder transaction then precise rule must be introduce.

Problems:

- the sequencer could still censor or delay transactions;
- the sequencer could leak information to external bots;
- in systems with a single sequencer, a strong level of trust is still required

PREVENTION/REDUCTION: FAIR ORDERING POLICIES

Single Sequencer

<u>First Come First Serve policy:</u>

- transactions are ordered according to their arrival time;
- whoever arrives first is processed first.

New problems emerge:

- Latency wars
- Spam attacks

PREVENTION/REDUCTION: FAIR ORDERING POLICIES

TimeBoost

Each transaction receives a score calculated using:

- a timestamp
- an economic bid

$$S(t_i, b_i) = \pi(b_i) - t_i$$

t_i = transaction arrival timestamp

b_i = bid (the tip paid by the user to gain priority)

$\pi(b_i)$ = bid priority function

PREVENTION/REDUCTION: FAIR ORDERING POLICIES

Decentralized Sequencing

Several nodes collaborate to decide transaction ordering instead of relying on a single operator.

Themis and Wendy: temporary leaders or synchronized timestamp across nodes

Require significant coordination and increase complexity and network overhead.

PREVENTION / REDUCTION: PRIVACY-PRESERVING METHODS

Transaction contents are hidden until the ordering has already been decided.

Threshold Encryption
- Transactions are encrypted
- Decryption key is distributed among multiple parties

Delay Encryption
- The transaction remains encrypted for a certain period of time
- Can only be obtained after solving a computational problem

Trusted Execution Environments
- Transactions are processed inside protected enclaves
- Published only after execution

PREVENTION / REDUCTION: SMART CONTRACT SOLUTIONS

Reduce the possibility of front-running directly inside applications.

Two main families:

- Entirely **on-chain** and remains inside smart contracts.
- **Off-chain** components connected to the blockchain ecosystem. Some important example are CoW Swap and Fair MM.

SIDE-EFFECT REDUCTION: PROPOSER-BUILDER SEPARATION

WHO BUILDS THE BLOCK

WHO PROPOSES IT TO THE NETWORK

The most important implementation today is **MEV-Boost**

In this model:

- searchers identify MEV opportunities;
- builders construct blocks;
- relays verify blocks;
- validators choose which block to propose.

✓ Reduces the power of individual validators

✗ Introduces new centralization risks around relays and builders

mev-boost

CONCLUSION

APPROACH	PROS	CONS
Fair Ordering Policies	Improves fairness and reduces transaction manipulation	Coordination overhead and latency issues
Privacy-Preserving Methods	Protects transaction privacy	High cryptographic complexity, increased latency and scalability challenges
Smart Contract-Level	No protocol changes required	Limited to specific applications
Proposer-Builder Separation	Improves scalability and efficiency and reduces validator power	New centralization risks (builders/relays) and higher ecosystem complexity

MEV cannot be completely eliminated, because it emerges directly from the structure of blockchains based on transaction ordering.